

Figure 10-16 Transmitter with radial velocity u_r approaching a stationary receiver.

with frequency f_t (**Fig. 10-16**). At a distance R from the source, the electric field of the radiated wave is given by

$$E(R) = E_0 e^{j(\omega_t t - kR)} = E_0 e^{j\phi}, \qquad (10.29)$$

where E_0 is the wave's magnitude, $\omega_t = 2\pi f_t$, and $k = 2\pi/\lambda_t$ with λ_t being the wavelength of the transmitted wave. The magnitude depends on the distance R and the gain of the source antenna, but it is not of concern as far as the Doppler effect is concerned. The quantity

$$\phi = \omega_t t - kR = 2\pi f_t t - \frac{2\pi}{\lambda_t} R \qquad (10.30)$$

is the phase of the radiated wave relative to its phase at $R = 0$ and reference time $t = 0$. If the source is moving toward the receiver, as in **Fig. 10-16**, or vice versa, at a radial velocity u_r, then

$$R = R_0 - u_r t, \qquad (10.31)$$

where R_0 is the distance between the source and the receiver at $t = 0$. Hence,

$$\phi = 2\pi f_t t - \frac{2\pi}{\lambda_t}(R_0 - u_r t). \qquad (10.32)$$

This is the phase of the signal detected by the receiver. The frequency of a wave is defined as the time derivative of the phase ϕ divided by 2π. Thus,

$$f_r = \frac{1}{2\pi}\frac{d\phi}{dt} = f_t + \frac{u_r}{\lambda_t}. \qquad (10.33)$$

Comparison of Eq. (10.33) with Eq. (10.28) leads to $f_d = u_r/\lambda_t$. For radar, the Doppler shift happens twice: once for the wave from the radar to the target and again for the wave reflected by the target back to the radar. Hence, $f_d = 2u_r/\lambda_t$. The dependence of f_d on direction is given by the dot product of the velocity and range unit vectors, which leads to

$$f_d = -2\frac{u_r}{\lambda_t} = -\frac{2u}{\lambda_t}\cos\theta, \qquad (10.34)$$

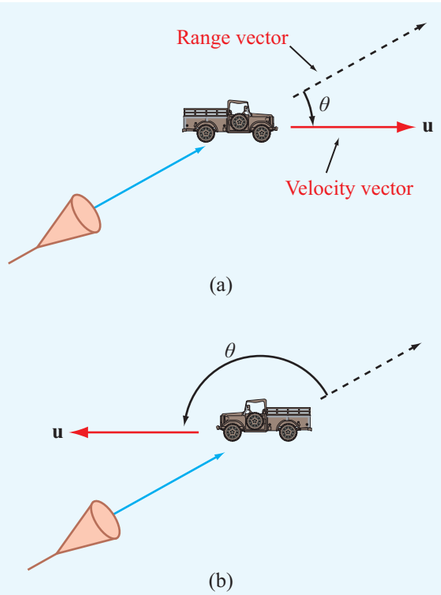


Figure 10-17 The Doppler frequency shift is negative for a receding target ($0 \le \theta \le 90°$), as in (a), and positive for an approaching target ($90° \le \theta \le 180°$), as in (b).

where u_r is the radial velocity component of u and θ is the angle between the range vector and the velocity vector (**Fig. 10-17**) with the direction of the range vector defined to be *from* the radar *to* the target. For a receding target (relative to the radar), $0 \le \theta \le 90°$, and for an approaching target, $90° \le \theta \le 180°$.

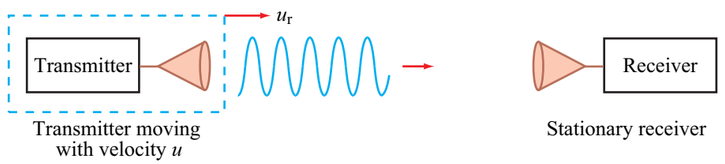


图 10-16 发射器以径向速度 u_r 接近静止接收器。

频率为 f_t (10-16)。在距源的距离 R 处，辐射波的电场由下式给出

$$E(R) = E_0 e^{j(\omega_t t - kR)} = E_0 e^{j\phi}, \qquad (10.29)$$

其中 E_0 是波的幅度，$\omega_t = 2\pi f_t$ 和 $k = 2\pi/\lambda_t$，其中 λ_t 是传输波的波长。幅度取决于距离 R 和源天线的增益，但就多普勒效应而言，它并不重要。数量

$$\phi = \omega_t t - kR = 2\pi f_t t - \frac{2\pi}{\lambda_t} R \qquad (10.30)$$

是辐射波的相位相对于其在 $R = 0$ 和参考时间 $t = 0$ 时的相位。如果源以径向速度 u_r 向接收器移动，如图 10-16 所示，或反之亦然，则

$$R = R_0 - u_r t, \qquad (10.31)$$

其中 R_0 是 $t = 0$ 时源与接收器之间的距离。因此，

$$\phi = 2\pi f_t t - \frac{2\pi}{\lambda_t}(R_0 - u_r t). \qquad (10.32)$$

这是接收器检测到的信号的相位。波的频率定义为相位 ϕ 除以 2π 的时间导数。因此，

$$f_r = \frac{1}{2\pi}\frac{d\phi}{dt} = f_t + \frac{u_r}{\lambda_t}. \qquad (10.33)$$

等式的比较（10.33）与方程。(10.28) 导致 $f_d = u_r/\lambda_t$。对于雷达，多普勒频移发生两次：一次是波从雷达到目标，另一次是波被目标反射回雷达。因此，$f_d = 2u_r/\lambda_t$。f_d 对方向的依赖性由速度和距离单位向量的点积给出，这导致

$$f_d = -2\frac{u_r}{\lambda_t} = -\frac{2u}{\lambda_t}\cos\theta, \qquad (10.34)$$

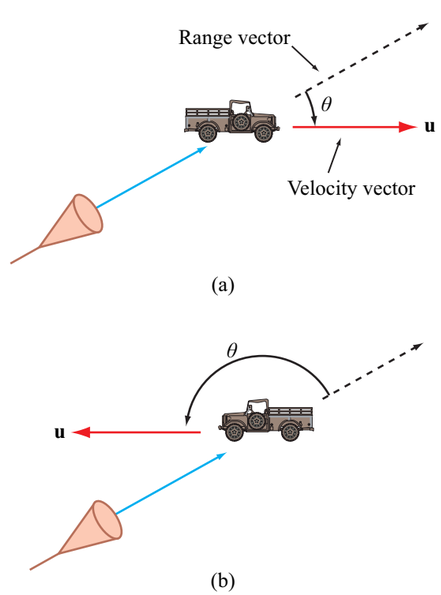


图 10-17 后退目标的多普勒频移为负 ($0 \le \theta \le 90°$)，如 (a) 所示；对于接近的目标，多普勒频移为正 ($90° \le \theta \le 180°$)，如 (b) 所示。

其中u_r是u的径向速度分量，θ是距离矢量和速度矢量之间的角度（图10-17），距离矢量的方向定义为从雷达到目标。对于后退目标（相对于雷达），$0 \le \theta \le 90°$，对于接近目标，$90° \le \theta \le 180°$。

10-8 Monopulse Radar

On the basis of information extracted from the echo due to a single pulse, a ***monopulse radar*** can track the direction of a target with an angular accuracy equal to a fraction of its antenna beamwidth. To track a target in both elevation and azimuth, a monopulse radar uses an antenna (such as a parabolic dish) with four separate small horns at its focal point (**Fig. 10-18**). Monopulse systems are of two types. The first is called ***amplitude-comparison monopulse*** because the tracking information is extracted from the amplitudes of the echoes received by the four horns, and the second is called ***phase-comparison monopulse*** because it relies on the phases of the received signals. We shall limit our present discussion to the amplitude-comparison scheme.

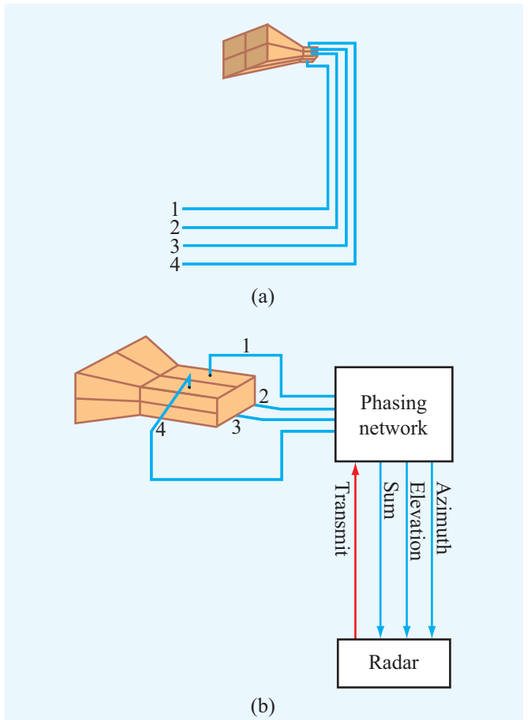


Figure 10-18 Antenna feeding arrangement for an amplitude-comparison monopulse radar: (a) feed horns and (b) connection to phasing network.

Individually, each horn would produce its own beam with the four beams pointing in slightly different directions. **Figure 10-19** shows the beams of two adjacent horns. The basic principle of the amplitude-comparison monopulse is to measure the amplitudes of the echo signals received through the two beams and then apply the difference between them to repoint the antenna boresight direction toward the target. Using computer-controlled phase shifters, the ***phasing network*** shown in **Fig. 10-18** can combine the signal delivered to the four-element horn array by the transmitter or by the echo signals received by them in different ways. Upon transmission, the network excites all four feeds in phase, thereby producing a single main beam called the ***sum beam***. The phasing network uses special microwave devices that allow it to provide the desired functionality during both the transmit and receive modes. Its equivalent functionality is described by the circuits shown in **Fig. 10-20**. During the receive period, the phasing network uses power dividers, power combiners, and phase shifters to generate three different output channels. One of these is the sum channel, corresponding to adding all four horns in phase, and its radiation pattern is depicted in **Fig. 10-21(a)**. The second channel, called the ***elevation-difference channel***, is obtained by first adding the outputs of the top-right and top-left horns (**Fig. 10-20(b)**), then adding the outputs of the bottom-right and bottom-left horns, and finally subtracting the second sum from the first. The subtraction process is accomplished by adding a 180° phase shifter in the path of the second sum before adding it to the first sum. The beam pattern of the elevation-difference channel is shown in **Fig. 10-21(b)**. If the observed target is centered between the two elevation beams, the receiver echoes will have the same strength for both beams, thereby producing a zero output from the elevation-difference channel. If it is not, the amplitude of the elevation-difference channel will be proportional to the angular deviation of the target from the boresight direction, and its sign will denote the direction of the deviation. The third channel (not shown in **Fig. 10-20**) is the ***azimuth-difference channel***, and it is

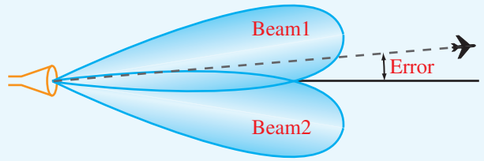


Figure 10-19 A target observed by two overlapping beams of a monopulse radar.

10-8 单脉冲雷达

基于从单个脉冲的回波中提取的信息，单脉冲雷达可以跟踪目标的方向，其角度精度等于其天线波束宽度的一部分。为了在仰角和方位角上跟踪目标，单脉冲雷达使用在其焦点处带有四个独立小喇叭的天线（例如抛物面天线）（图10-18）。单脉冲系统有两种类型。第一种称为幅度比较单脉冲，因为跟踪信息是从四个喇叭接收到的回波的幅度中提取的；第二种称为相位比较单脉冲，因为它依赖于接收信号的相位。我们将把目前的讨论限制在幅度比较方案上。

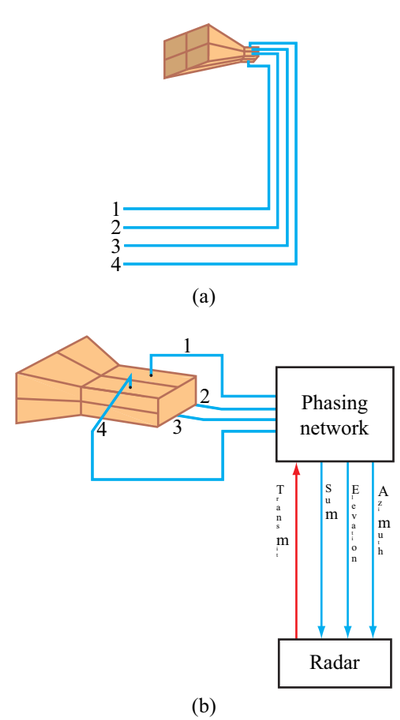


图 10-18 幅度比较单脉冲雷达的天线馈电布置：(a) 馈电喇叭和 (b) 与定相网络的连　接。

每个喇叭都会单独产生自己的波束，四个波束指向略有不同的方向。图10-19显示了两个相邻喇叭的波束。幅度比较单脉冲的基本原理是测量通过两个波束接收到的回波信号的幅度，然后应用它们之间的差值将天线视轴方向重新指向目标。使用计算机控制的移相器，图10-18所示的定相网络可以将发射器传递到四元件喇叭阵列的信号或它们以不同方式接收到的回波信号组合起来。传输时，网络同相激励所有四个馈源，从而产生称为和波束的单个主波束。定相网络使用特殊的微波设备，使其能够在发射和接收模式下提供所需的功能。其等效功能由图 10-20 所示的电路描述。在接收期间，定相网络使用功率分配器、功率组合器和移相器来生成三个不同的输出通道。其中之一是求和通道，相当于将所有四个喇叭同相相加，其辐射图如图 10-21(a) 所示。第二个通道称为俯仰差通道，首先将右上角和左上角的输出相加（图 10-20（b）），然后将右下角和左下角的输出相加，最后从第一个和中减去第二个和。减法过程是通过在将第二个和添加到第一个和之前在第二个和的路径中添加一个 180° 移相器来完成的。俯仰差通道的波束方向图如图10-21(b)所示。如果观察到的目标位于两个仰角波束之间的中心，则两个波束的接收器回波将具有相同的强度，从而从仰角差通道产生零输出。如果不是，则仰角通道的幅度将与目标相对视轴方向的角度偏差成正比，其符号表示偏差的方向。第三个通道（图10-20中未示出）是方位差通道，它是

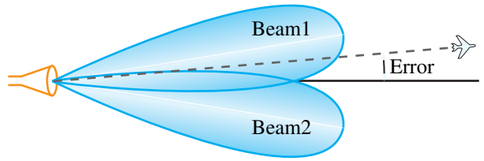


图10-19 单脉冲雷达的两个重叠波束观测到的目标。

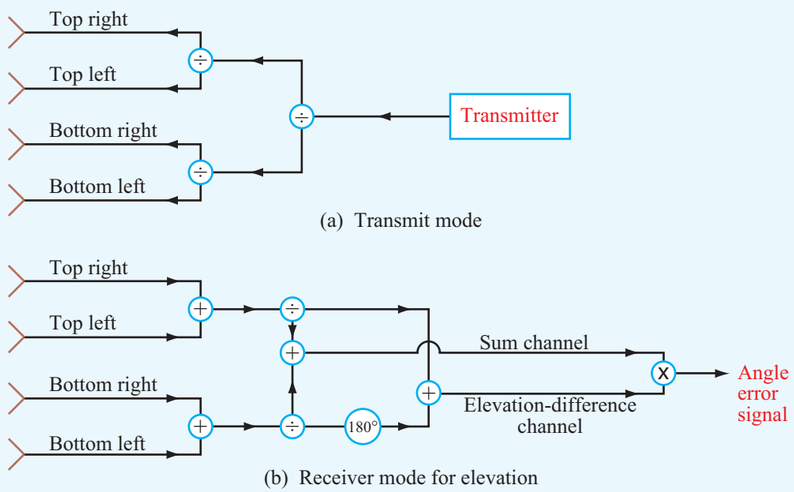


Figure 10-20 Functionality of the phasing network in (a) the transmit mode and (b) the receive mode for the elevation-difference channel.

accomplished through a similar process that generates a beam corresponding to the difference between the sum of the two right horns and the sum of the two left horns.

In practice, the output of the difference channel is multiplied by the output of the sum channel to increase the strength of the difference signal and to provide a phase reference for extracting the sign of the angle. This product, called the ***angle error signal***, is displayed in **Fig. 10-21(c)** as a function of the angle error. The error signal activates a servo-control system to reposition the antenna direction. By applying a similar procedure along the azimuth direction using the product of the azimuth-difference channel and the sum channel, a monopulse radar provides automatic tracking in both directions. The range to the target is obtained by measuring the round-trip delay of the signal.

Concept Question 10-5: How is the PRF related to unambiguous range?

Concept Question 10-6: Explain how the false-alarm probability and the detection probability are related to the noise level of the receiver.

Concept Question 10-7: In terms of the geometry shown in **Fig. 10-17**, when is the Doppler shift a maximum?

Concept Question 10-8: What is the principle of the monopulse radar?

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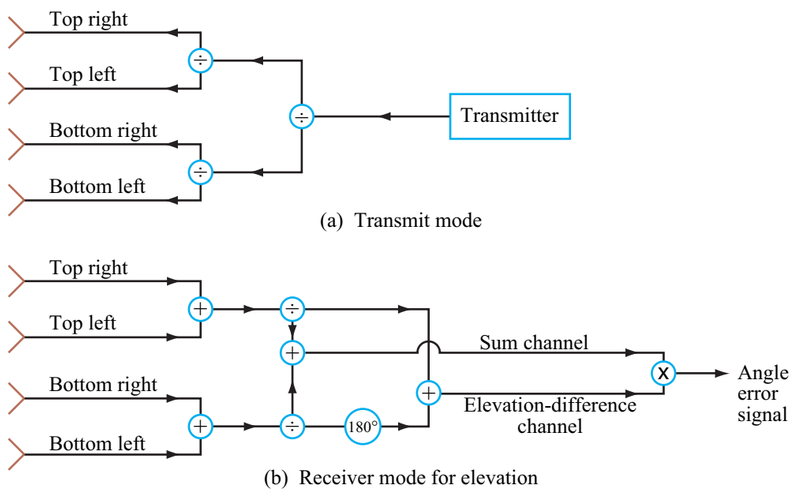


图 10-20 功能 (a) 发射模式和 (b) 接收模式下的定相网络 m 高差通道的颂歌。

通过类似的过程来完成，该过程生成与两个右喇叭的总和与两个左喇叭的总和之间的差相对应的光束。

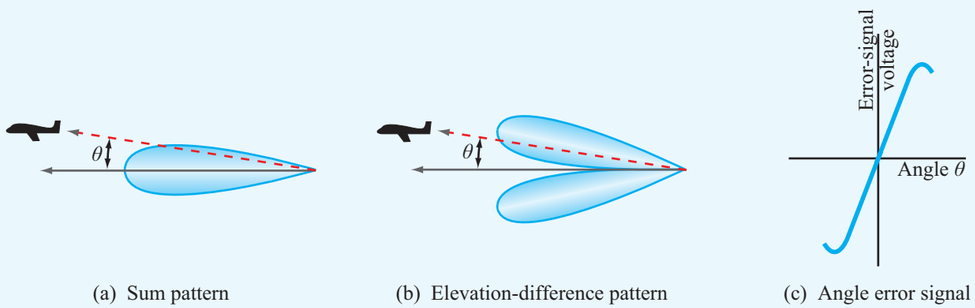
实际上，差通道的输出乘以和通道的输出，以增加差信号的强度并为提取角度符号提供相位参考。该乘积称为角度误差信号，作为角度误差的函数显示在图 10-21(c) 中。误差信号激活伺服控制系统以重新定位天线方向。通过使用方位差通道与和通道的乘积沿着方位角方向应用类似的过程，单脉冲雷达提供两个方向的自动跟踪。通过测量信号的往返延迟来获得到目标的距离。

概念问题 10-5：PRF 与明确范围有何关系？

概念问题 10-6：解释误报概率和检测概率与接收器噪声水平的关系。

概念问题10-7：根据图10-17所示的几何形状，多普勒频移何时达到最大值？

概念题10-8：单脉冲雷达的原理是什么？



(a) Sum pattern (b) Elevation-difference pattern (c) Angle error signal

Figure 10-21 Monopulse antenna (a) sum pattern, (b) elevation-difference pattern, and (c) angle error signal.

Chapter 10 Summary

Concepts

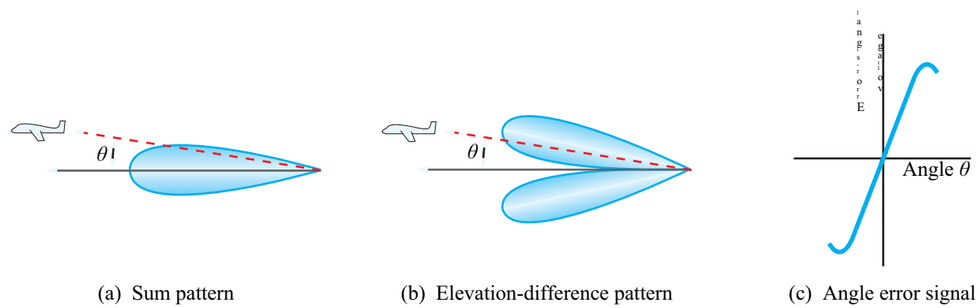
- Three equally spaced satellites in geostationary orbit can provide coverage of most of Earth's surface.
- The use of polarization diversity makes it possible to double the number of channels per unit bandwidth carried by a satellite repeater.
- A satellite antenna system is designed to produce beams tailored to match the areas served by the satellite. Antenna arrays are particularly suitable for this purpose.
- A radar is an electromagnetic sensor that illuminates a region of space and then measures the echoes due to reflecting objects. From the echoes, information can be extracted about the range of a target, its radial velocity, direction of motion, and other characteristics.
- Due to the random nature of receiver noise, target detection is a statistical process characterized by detection and false-alarm probabilities.
- A moving object produces a Doppler frequency shift proportional to the radial velocity of the object (relative to the radar) and inversely proportional to λ.
- A monopulse radar uses multiple beams to track the direction of a target with an angular accuracy equal to a fraction of its antenna beamwidth.

Important Terms

Provide definitions or explain the meaning of the following terms:

atmospheric transmissivity Υ
azimuth resolution
bistatic radar
circulator
detection probability
Doppler frequency shift f_d
duplexer
Explorer I
false-alarm probability
FDMA
geostationary orbit
interpulse period T_p
lidar
matched filter
maximum detectable range R_{max}
monopulse radar
monostatic radar
multiplexer
polarization diversity
pulse length τ
pulse repetition frequency (PRF) f_p
radar
radar cross section σ_t
radar equation
radial velocity u_r
range resolution
Score
signal-to-noise ratio
Sputnik I
sum and difference channels
synchronizer
system noise temperature
threshold detection level
transponder
unambiguous range R_u
uplink and downlink

10-8 MONOPULSE RADAR 449



(a) Sum pattern (b) Elevation-difference pattern (c) Angle error signal

图 10-21 单脉冲天线 (a) 和图，(b) 仰角差图，(c) 角度误差信号。

第10章总结

概念

- 地球静止轨道上的三颗等距卫星可以覆盖地球表面的大部分区域。
- 极化分集的使用使得卫星中继器所承载的每单位带宽的信道数量加倍成为可能。
- 卫星天线系统旨在产生适合卫星服务区域的波束。天线阵列特别适合此目的。
- 雷达是一种电磁传感器，可以照亮空间区域，然后测量反射物体产生的回波。从回波中可以得到信息提取目标的范围、其径向速度、运动方向和其他特征。
- 由于接收机噪声的随机性，目标检测是一个以检测和误报概率为特征的统计过程。
- 移动物体产生的多普勒频移与物体的径向速度（相对于雷达）成正比，与 λ 成反比。
- 单脉冲雷达使用多个波束来跟踪目标的方向，其角度精度等于其天线波束宽度的一部分。

重要术语 提供以下术语的定义或解释其含义：

大气透射率 Υ 方位分辨率 双基地雷达循环器探测概率 多普勒频移 f_d 双工器 E xplorer I 虚警概率 FDMA 对地静止轨道脉冲间周期 T_p
激光雷达匹配滤波器 最大可探测距离 R_{max} 单脉冲雷达 单基地雷达多路复用器 极化分集 脉冲长度 τ 脉冲重复频率 (PRF) f_p 雷达 雷达截面 σ_t 雷达方程
径向速度 u_r 距离分辨率 得分 信噪比 Sputnik I 和差通道 同步器系统 噪声温度阈值 检测电平 转发器 明确距离 R_u 上行链路和下行链路

Mathematical and Physical Models

Satellite Communication Systems

Radius of geostationary orbit

$$R_0 = \left(\frac{GM_eT^2}{4\pi^2}\right)^{1/3}$$

Received power

$$P_{ri} = \Upsilon(\theta)\, P_r = \Upsilon(\theta)\, P_tG_tG_r\left(\frac{\lambda}{4\pi R}\right)^2$$

Noise power

$$P_{ni} = KT_{sys}B$$

Signal-to-noise ratio

$$S_n = \frac{P_{ri}}{P_{ni}} = \frac{\Upsilon(\theta)\, P_tG_tG_r}{KT_{sys}B}\left(\frac{\lambda}{4\pi R}\right)^2$$

Radar Sensors

Unambiguous range

$$R_u = \frac{cT_p}{2} = \frac{c}{2f_p}$$

Range resolution

$$\Delta R = R_2 - R_1 = c\tau/2$$

Azimuth resolution

$$\Delta x = \beta R$$

Radar equation

$$P_r = \frac{P_tG^2\lambda^2\sigma_t}{(4\pi)^3R^4}$$

Doppler frequency shift

$$f_d = -2\frac{u_r}{\lambda_t} = -\frac{2u}{\lambda_t}\cos\theta$$

PROBLEMS

Sections 10-1 to 10-4: Satellite Communication Systems

*10.1 A remote sensing satellite is in circular orbit around Earth at an altitude of 1,100 km above Earth's surface. What is its orbital period?

10.2 A transponder with a bandwidth of 400 MHz uses polarization diversity. If the bandwidth allocated to transmit a single telephone channel is 4 kHz, how many telephone channels can be carried by the transponder?

*10.3 Repeat Problem 10.2 for TV channels with each requiring a bandwidth of 6 MHz.

10.4 A geostationary satellite is at a distance of 40,000 km from a ground receiving station. The satellite transmitting antenna is a circular aperture with a 1 m diameter, and the ground station uses a parabolic dish antenna with an effective diameter of 20 cm. If the satellite transmits 1 kW of power at 12 GHz and the ground receiver is characterized by a system noise temperature of 1,000 K, what would be the signal-to-noise ratio of a received TV signal with a bandwidth of 6 MHz? The antennas and the atmosphere may be assumed lossless.

Sections 10-5 to 10-8: Radar Sensors

*10.5 A collision-avoidance automotive radar is designed to detect the presence of vehicles up to a range of 0.5 km. What is the maximum usable PRF?

10.6 A 10 GHz weather radar uses a 15-cm diameter lossless antenna. At a distance of 1 km, what are the dimensions of the volume resolvable by the radar if the pulse length is 1 μs?

*10.7 A radar system is characterized by the following parameters: $P_t = 1$ kW, $\tau = 0.1$ μs, $G = 30$ dB, $\lambda = 3$ cm, and $T_{sys} = 1{,}500$ K. The radar cross section of a car is typically 5 m^2. How far away can the car be and remain detectable by the radar with a minimum signal-to-noise ratio of 13 dB?

10.8 A 3-cm wavelength radar is located at the origin of an x–y coordinate system. A car located at $x = 100$ m and $y = 200$ m is heading east (x direction) at a speed of 120 km/hr. What is the Doppler frequency measured by the radar?

* Answer(s) available in Appendix E.

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数学与物理模型

卫星通信系统

地球静止轨道半径

$$R_0 = \left(\frac{GM_eT^2}{4\pi^2}\right)^{1/3}$$

接收功率

$$P_{ri} = \Upsilon(\theta)\, P_r = \Upsilon(\theta)\, P_tG_tG_r\left(\frac{\lambda}{4\pi R}\right)^2$$

噪声功率

$$P_{ni} = KT_{sys}B$$

信噪比

$$S_n = \frac{P_{ri}}{P_{ni}} = \frac{\Upsilon(\theta)\, P_tG_tG_r}{KT_{sys}B}\left(\frac{\lambda}{4\pi R}\right)^2$$

雷达传感器

最大不模糊距离

$$R_u = \frac{cT_p}{2} = \frac{c}{2f_p}$$

距离分辨率

$$\Delta R = R_2 - R_1 = c\tau/2$$

方位分辨率

$$\Delta x = \beta R$$

雷达方程

$$P_r = \frac{P_tG^2\lambda^2\sigma_t}{(4\pi)^3R^4}$$

多普勒频移

$$f_d = -2\frac{u_r}{\lambda_t} = -\frac{2u}{\lambda_t}\cos\theta$$

问题

第10-1至10-4节 卫星通信系统

*10.1 一颗遥感卫星在距地面1100 km的圆轨道上绕地球运行。求其轨道周期。

10.2 带宽为400 MHz的转发器采用极化分集。若每个电话信道占用4 kHz带宽，该转发器能承载多少个电话信道?

*10.3 若每个电视信道需要6 MHz带宽，重做习题10.2。

10.4 一颗地球静止卫星距地面接收站40000 km。卫星发射天线为直径1 m的圆形口径，地面站抛物面天线的有效直径为20 cm。卫星在12 GHz频率上发射1 kW功率，接收机系统噪声温度为1000 K。若接收的电视信号带宽为6 MHz，求信噪比。假设天线和大气均无损。

第10-5至10-8节 雷达传感器

*10.5 一部汽车防碰撞雷达需要探测最远0.5 km处的车辆。求可用的最大脉冲重复频率（PRF）。

10.6 一部10 GHz气象雷达采用直径15 cm的无损天线。若脉冲宽度为1 μs，求距雷达1 km处可分辨体积的各方向尺寸。

*10.7 某雷达的参数为 P_t=1 kW、τ=0.1 μs、G=30 dB、λ=3 cm、T_{sys}=1500 K。汽车的典型雷达散射截面为5 m^2。若探测所需的最小信噪比为13 dB，求仍能探测到汽车的最大距离。

10.8 一部波长为3 cm的雷达位于 x-y 坐标系原点。一辆汽车位于 x=100 m、y=200 m处，正以120 km/h的速度向东（+x方向）行驶。求雷达测得的多普勒频移。

* 标记的习题在附录 E 提供答案。